

## 1 Probability

We begin by collecting the probabilistic results on which every later section relies. The notation  $\bar{Y}_n = \frac{1}{n} \sum_{i=1}^n Y_i$  denotes the sample mean throughout.

### 1.1 Modes of Convergence

**In probability:**  $Y_n \xrightarrow{\mathbb{P}} Y$  if  $\mathbb{P}(|Y_n - Y| > \epsilon) \rightarrow 0$  for every  $\epsilon > 0$ .

**In distribution (law):**  $Y_n \xrightarrow{d} Y$  if  $F_{Y_n}(x) \rightarrow F_Y(x)$  at every continuity point  $x$  of  $F_Y$ .

**In quadratic mean ( $L^2$ ):**  $\mathbb{E}[(Y_n - Y)^2] \rightarrow 0$ . Convergence in  $L^2$  implies convergence in probability, which in turn implies convergence in distribution.

**Continuous Mapping Theorem:** If  $g$  is continuous, then  $Y_n \xrightarrow{\mathbb{P}} Y \Rightarrow g(Y_n) \xrightarrow{\mathbb{P}} g(Y)$ , and  $Y_n \xrightarrow{d} Y \Rightarrow g(Y_n) \xrightarrow{d} g(Y)$ .

**Slutsky's Theorem:** If  $Y_n \xrightarrow{d} Y$  and  $K_n \xrightarrow{\mathbb{P}} k$  with  $k$  a *constant*, then

$$K_n + Y_n \xrightarrow{d} k + Y, \quad K_n Y_n \xrightarrow{d} kY, \quad Y_n/K_n \xrightarrow{d} Y/k \quad (k \neq 0)$$

The limit of  $K_n$  must be deterministic; the result can fail otherwise.

**Weak Law of Large Numbers:** If  $Y_i$  are iid with  $\mathbb{E}|Y_1| < \infty$ , then  $\bar{Y}_n \xrightarrow{\mathbb{P}} \mathbb{E}(Y_1)$ .

**Central Limit Theorem:** If  $Y_i$  are iid with  $\text{Var}(Y_1) < \infty$ , then

$$\frac{\bar{Y}_n - \mathbb{E}(Y_1)}{\sqrt{\text{Var}(\bar{Y}_n)}} = \sqrt{n} \frac{\bar{Y}_n - \mathbb{E}(Y_1)}{\sqrt{\text{Var}(Y_1)}} \xrightarrow{d} \mathcal{N}(0, 1)$$

**Delta Method:** If  $a_n(Y_n - b) \xrightarrow{d} Z$  with  $a_n \rightarrow \infty$  and  $g$  is differentiable at  $b$ , then  $a_n(g(Y_n) - g(b)) \xrightarrow{d} g'(b)Z$ . In particular,  $\sqrt{n}(Y_n - b) \xrightarrow{d} \mathcal{N}(0, s^2)$  gives  $\sqrt{n}(g(Y_n) - g(b)) \xrightarrow{d} \mathcal{N}(0, g'(b)^2 s^2)$ .

**Sample Variance:** If  $\mathbb{E}(Y_1^4) < \infty$ ,  $v = \text{Var}(Y_1)$  and  $\kappa = \mathbb{E}[(Y_1 - \mathbb{E}Y_1)^4]/v^2$  is the kurtosis, then  $V_n = \frac{1}{n} \sum (Y_i - \bar{Y}_n)^2$  satisfies

$$\sqrt{n} \frac{V_n - v}{\sqrt{v^2(\kappa - 1)}} \xrightarrow{d} \mathcal{N}(0, 1)$$

For Gaussian data  $\kappa = 3$ , so the asymptotic variance is  $2v^2$ .

### 1.2 Gaussian Random Variables

- If  $X \sim \mathcal{N}(m, \sigma^2)$  then  $(X - m)/\sigma \sim \mathcal{N}(0, 1)$

- The standard normal CDF  $\Phi$  satisfies  $\Phi(x) = 1 - \Phi(-x)$ , and its quantiles satisfy  $q_\alpha = -q_{1-\alpha}$

- Independent  $X_i \sim \mathcal{N}(m_i, \sigma_i^2)$  give  $\sum a_i X_i \sim \mathcal{N}(\sum a_i m_i, \sum a_i^2 \sigma_i^2)$

- For  $Z \sim \mathcal{N}(0, 1)$ : odd moments vanish and  $\mathbb{E}(Z^{2k}) = (2k - 1)(2k - 3) \cdots 1$ , so  $\mathbb{E}(Z^4) = 3$

**Useful Quantiles:**  $q_{0.95} \approx 1.645$ ,  $q_{0.975} \approx 1.960$ ,  $q_{0.995} \approx 2.576$ .

### 1.3 Random Vectors

For a random vector  $\mathbf{Z} \in \mathbb{R}^p$  we define  $\mathbb{E}(\mathbf{Z})$  componentwise and

$$\text{Cov}(\mathbf{Z}) = \mathbb{E}[(\mathbf{Z} - \mathbb{E}\mathbf{Z})(\mathbf{Z} - \mathbb{E}\mathbf{Z})^T]$$

which is symmetric and positive semi-definite. For a deterministic matrix  $A$  and vector  $\mathbf{b}$ ,

$$\mathbb{E}(A\mathbf{Z} + \mathbf{b}) = A\mathbb{E}(\mathbf{Z}) + \mathbf{b}, \quad \text{Cov}(A\mathbf{Z} + \mathbf{b}) = A\text{Cov}(\mathbf{Z})A^T$$

**Gaussian Vectors:**  $\mathbf{Z}$  is Gaussian if *every* linear combination  $\mathbf{a}^T \mathbf{Z}$  is normal. Normal marginals alone are not enough. If  $\mathbf{Z} \sim \mathcal{N}_p(\boldsymbol{\mu}, \Sigma)$  then  $A\mathbf{Z} + \mathbf{b} \sim \mathcal{N}_m(A\boldsymbol{\mu} + \mathbf{b}, A\Sigma A^T)$ .

**Uncorrelated vs Independent:** Independence always implies zero covariance. The converse holds only for jointly Gaussian vectors: if  $\Sigma$  is diagonal, the components are independent.

**Quadratic Form:** If  $\mathbf{Y} \sim \mathcal{N}_n(\boldsymbol{\mu}, \Sigma)$  with  $\Sigma$  invertible, then  $(\mathbf{Y} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{Y} - \boldsymbol{\mu}) \sim \chi_n^2$ .

### 1.4 Common Distributions

| Distribution             | Density / definition                                                      | Mean, Variance                                    |
|--------------------------|---------------------------------------------------------------------------|---------------------------------------------------|
| Gamma( $\alpha, \beta$ ) | $\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$ , $x > 0$ | $\frac{\alpha}{\beta}$ , $\frac{\alpha}{\beta^2}$ |
| Exp( $\beta$ )           | Gamma(1, $\beta$ ): $\beta e^{-\beta x}$                                  | $\frac{1}{\beta}$ , $\frac{1}{\beta^2}$           |
| $\chi_k^2$               | Gamma( $k/2, 1/2$ ), or $\sum_{i=1}^k Z_i^2$                              | $k$ , $2k$                                        |
| $t_k$                    | $Z/\sqrt{X/k}$ , $Z \perp X \sim \chi_k^2$                                | 0, $\frac{k}{k-2}$ ( $k > 2$ )                    |
| $F_{k_1, k_2}$           | $\frac{X_1/k_1}{X_2/k_2}$ , independent $\chi^2$                          |                                                   |
| Beta( $a, b$ )           | $\frac{x^{a-1}(1-x)^{b-1}}{B(a, b)}$ , $0 < x < 1$                        | $\frac{a}{a+b}$                                   |

- $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$ , with  $\Gamma(n) = (n-1)!$ ; here  $\beta$  is a rate, so check whether software expects a rate or a scale  $1/\beta$
- Exponential:  $F(x) = 1 - e^{-\beta x}$ , quantile  $-\ln(1-u)/\beta$ , and the minimum of independent  $\text{Exp}(\beta_i)$  is  $\text{Exp}(\sum \beta_i)$
- The  $t_k$  density is symmetric with heavier tails than  $\mathcal{N}(0, 1)$  and tends to it as  $k \rightarrow \infty$
- Sum of  $n$  iid  $\text{Exp}(\beta)$  variables is  $\text{Gamma}(n, \beta)$

## 2 Point Estimation

### 2.1 Statistical Model

We observe data  $(x_1, \dots, x_n)$  and treat them as a realisation of random variables  $(X_1, \dots, X_n)$  on some probability space. In a **parametric model** their joint law is known up to an unknown parameter  $\theta \in \Theta \subseteq \mathbb{R}^d$ . Most of what follows assumes an iid sample.

**Illustrations:** Loan default indicators can be modelled as iid Bernoulli( $p$ ) with  $p \in (0, 1)$  unknown ( $d = 1$ ). Daily log returns can be modelled as iid  $\mathcal{N}(\mu, v)$  with  $\theta = (\mu, v)$  unknown ( $d = 2$ ).

**Typical Tasks:** Give a single value for  $\theta$  (*point estimation*), a set containing  $\theta$  with high probability (*confidence region*), or a rule to choose between two claims about  $\theta$  (*hypothesis testing*).

**Estimator:** Any measurable function  $h(X_1, \dots, X_n)$  that does *not* depend on  $\theta$ . For example  $\sum X_i$ ,  $X_1$  and  $(X_1 + X_n)/2$  are estimators, whereas  $X_1 - \theta$  is not.

### 2.2 Maximum Likelihood

**Likelihood:** In the discrete case  $L(x; \theta) = \mathbb{P}_\theta(X_1 = x_1, \dots, X_n = x_n)$ . In the continuous case this probability is zero, so we use the joint density instead,  $L(x; \theta) = f_{(X_1, \dots, X_n)}(x_1, \dots, x_n; \theta)$ . Under independence and identical distribution both reduce to

$$L(x; \theta) = \prod_{i=1}^n f_\theta(x_i)$$

where  $f_\theta$  is the pmf or pdf of  $X_1$ . Measure theoretically, both are densities with respect to a dominating measure (counting or Lebesgue).

**MLE:** The maximiser  $\hat{\theta}_n = h^*(X_1, \dots, X_n)$  of  $\theta \mapsto L(x; \theta)$  over  $\Theta$ , when it exists. Since log is strictly increasing, we usually maximise the **log likelihood**  $\ell(\theta) = \sum_i \log f_\theta(x_i)$ , solve the score equation  $\ell'(\theta) = 0$ , and check the second order condition (negative second derivative, or a negative definite Hessian).

**Bernoulli( $p$ ):** With  $f_p(x) = p^x(1-p)^{1-x}$ ,  $x \in \{0, 1\}$ ,

$$\ell(p) = \left( \sum x_i \right) \log p + \left( n - \sum x_i \right) \log(1-p) \Rightarrow \hat{p} = \bar{X}_n$$

**Normal( $\mu, v$ ):**

$$\ell(\mu, v) = -\frac{n}{2} \log v - \frac{n}{2} \log(2\pi) - \frac{1}{2v} \sum_{i=1}^n (x_i - \mu)^2$$

Setting  $\partial_\mu \ell = \partial_v \ell = 0$  gives

$$\hat{\mu} = \bar{X}_n, \quad \hat{v} = V_n = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

**When the Support Depends on  $\theta$ :** Calculus fails and we reason directly. For  $X_i \stackrel{\text{iid}}{\sim} \text{Uniform}(0, \theta)$ ,  $L(\theta) = \theta^{-n} \mathbb{1}_{\{\theta \geq \max_i x_i\}}$  is decreasing on its support, so  $\hat{\theta} = \max_i X_i$  (which is biased downwards).

**Invariance:** If  $\hat{\theta}$  is the MLE of  $\theta$ , then  $g(\hat{\theta})$  is the MLE of  $g(\theta)$ .

### 2.3 Properties of Estimators

**Unbiased:**  $\mathbb{E}_\theta(\hat{\theta}_n) = \theta$  for every  $\theta \in \Theta$ . The bias is  $\text{Bias}(\theta, \hat{\theta}) = \mathbb{E}_\theta(\hat{\theta}) - \theta$ . An estimator is **asymptotically unbiased** if the bias vanishes as  $n \rightarrow \infty$ .

**Consistent (Convergent):**  $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta$  for every  $\theta$ . By the WLLN,  $\bar{X}_n$  is consistent for  $\mathbb{E}(X_1)$ .

**Asymptotically Normal:** For consistent  $\hat{\theta}_n$ ,

$$\frac{\hat{\theta}_n - \theta}{\sqrt{\text{Var}(\hat{\theta}_n)}} \xrightarrow{d} \mathcal{N}(0, 1)$$

For the Bernoulli MLE, the CLT gives  $\sqrt{n}(\bar{X}_n - p)/\sqrt{p(1-p)} \xrightarrow{d} \mathcal{N}(0, 1)$ .

**Sample Mean and Sample Variance:** For an iid sample with mean  $\mu$  and variance  $\sigma^2$ :

| Estimator                                        | Expectation              | Comments                                        |
|--------------------------------------------------|--------------------------|-------------------------------------------------|
| $\bar{X}_n$                                      | $\mu$                    | Unbiased, consistent, $\text{Var} = \sigma^2/n$ |
| $V_n = \frac{1}{n} \sum (X_i - \bar{X}_n)^2$     | $\frac{n-1}{n} \sigma^2$ | Biased; asympt. unbiased, consistent            |
| $S_n^2 = \frac{1}{n-1} \sum (X_i - \bar{X}_n)^2$ | $\sigma^2$               | Unbiased, consistent                            |

Consistency of  $V_n$  and  $S_n^2$  follows from the WLLN applied to  $X_i$  and  $X_i^2$  with the continuous mapping theorem; asymptotic normality needs  $\mathbb{E}(X_1^4) < \infty$ . Hence the Gaussian MLE of the variance is biased, although it is consistent and asymptotically normal.

**Asymptotics of the MLE:** Under regularity conditions the MLE is consistent and  $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} \mathcal{N}(0, I(\theta)^{-1})$ , where  $I(\theta)$  is the Fisher information defined below.

### 2.4 Sufficiency

**Sufficient Estimator:**  $T(X)$  is sufficient for  $\theta$  if the conditional law of the sample given  $T(X)$  does not depend on  $\theta$ , i.e.  $\mathbb{E}_\theta[\mathbb{1}_{\{X \in B\}} | T(X)]$  is free of  $\theta$  for every event  $B$ . Once  $T$  is known, the data carry no further information about  $\theta$ .

**Factorisation Theorem (Fisher and Neyman):**  $T(X)$  is sufficient if and only if

$$L(x; \theta) = g_\theta(T(x)) H(x)$$

where  $H$  does not depend on  $\theta$ . Sufficient statistics are not unique (any one to one function of one is also sufficient).

**Standard Cases:**

- Bernoulli( $p$ ):  $L = p^{\sum x_i} (1-p)^{n-\sum x_i}$ , so  $\sum X_i$  is sufficient
- $\mathcal{N}(\mu, \sigma^2)$ :  $L \propto \sigma^{-n} \exp(-\frac{1}{2\sigma^2}(\sum x_i^2 - 2\mu \sum x_i + n\mu^2))$ , so  $(\sum X_i, \sum X_i^2)$  is sufficient
- Exp( $\theta$ ):  $L = \theta^n e^{-\theta \sum x_i}$ , so  $\sum X_i \sim \text{Gamma}(n, \theta)$  is sufficient

**Link with the MLE:** If the MLE is unique, it is a function of any sufficient statistic.

### 2.5 Decision Theory

**Loss and Risk:** A loss  $L(\theta, d) \geq 0$  penalises the decision  $d$  when the truth is  $\theta$ ; the common choice is the **quadratic loss**  $(\theta - d)^2$ . The **risk** of an estimator  $T$  is its average loss,

$$R(\theta, T) = \mathbb{E}_\theta \big[ L(\theta, T(X)) \big]$$

For the Bernoulli sample mean under quadratic loss,

$$R(p, \bar{X}_n) = \text{Var}_p(\bar{X}_n) = p(1 - p)/n.$$

**Bias Variance Decomposition:**

$$R(\theta, T) = \text{Var}_\theta \big( T(X) \big) + \text{Bias}(\theta, T)^2$$

**Admissibility:**  $T'$  **dominates**  $T$  if  $R(\theta, T') \leq R(\theta, T)$  for all  $\theta$ , with strict inequality for at least one  $\theta$ . A dominated estimator is **inadmissible**. In practice an inadmissible estimator may still be used if the better one is unknown or costly to compute.

**No Uniformly Best Estimator:** Risk functions of different estimators typically cross, so there is no estimator with smallest risk at every  $\theta$ . Two ways out are:

- Minimax:** choose  $T$  to minimise the worst case risk,  $\inf_T \sup_\theta R(\theta, T)$

- Restrict the class:** among *unbiased* estimators under quadratic loss, the risk equals the variance, and a uniformly best one may exist (the UMVUE)

### 2.6 Towards the UMVUE

**Rao Blackwell Theorem:** Let  $Y_1$  be sufficient and  $Y_2$  unbiased for  $\theta$ . Then  $\mathbb{E}_\theta[Y_2 \mid Y_1]$  is free of  $\theta$  (by sufficiency, so it is an estimator), and

$$\mathbb{E}_\theta \big[ \mathbb{E}(Y_2 \mid Y_1) \big] = \theta, \qquad \text{Var}_\theta \big( \mathbb{E}(Y_2 \mid Y_1) \big) \leq \text{Var}_\theta(Y_2)$$

with strict inequality unless  $Y_2$  is already a function of  $Y_1$ . Conditioning on a sufficient statistic never hurts, so the search for a best unbiased estimator can be restricted to functions of sufficient statistics.

**Completeness:** A sufficient  $Y$  is complete if  $\mathbb{E}_\theta[g(Y)] = 0$  for all  $\theta$  implies  $\mathbb{P}_\theta(g(Y) = 0) = 1$  for all  $\theta$ . Informally, the only unbiased estimator of zero built from  $Y$  is zero itself, so an unbiased estimator that is a function of  $Y$  is essentially unique.

**Lehmann Scheffé Theorem:** If  $Y_1$  is complete sufficient and  $h(Y_1)$  is unbiased for  $\theta$ , then  $h(Y_1)$  is the **UMVUE**:  $\text{Var}_\theta[h(Y_1)] \leq \text{Var}_\theta(Y_2)$  for every unbiased  $Y_2$  and every  $\theta$ . The proof combines Rao Blackwell ( $\mathbb{E}(Y_2 \mid Y_1)$  is unbiased with smaller variance) with completeness ( $\mathbb{E}(Y_2 \mid Y_1) = h(Y_1)$  almost surely).

#### 2.7 Cramér Rao Lower Bound

**Regularity:** We assume differentiation with respect to  $\theta$  can be exchanged with integration (or summation). This fails when the support depends on  $\theta$ , e.g.  $\text{Uniform}(0, \theta)$ .

**Score and Fisher Information:** The score  $\frac{\partial}{\partial \theta} \log f_\theta(X_1)$  has mean zero under regularity, and

$$I(\theta) = \mathbb{E}_\theta \bigg[ \bigg( \frac{\partial}{\partial \theta} \log f_\theta(X_1) \bigg)^2 \bigg] = -\mathbb{E}_\theta \bigg[ \frac{\partial^2}{\partial \theta^2} \log f_\theta(X_1) \bigg]$$

the second form requiring twice differentiability. For  $\theta \in \mathbb{R}^d$ ,  $I(\theta)$  is the  $d \times d$  matrix with entries  $\mathbb{E}_\theta[\partial_{\theta_j} \log f_\theta(X_1) \, \partial_{\theta_k} \log f_\theta(X_1)]$ .

**Inequality:** For any unbiased estimator  $Y$  based on an iid sample of size  $n$ ,

$$\text{Var}_\theta(Y) \geq \frac{1}{n \, I(\theta)}$$

The proof uses the zero mean of the score and the Cauchy Schwarz inequality.

**Efficiency:** An unbiased estimator attaining the bound is **efficient**, and is then necessarily the UMVUE. An efficient estimator need not exist. For Bernoulli( $p$ ),  $I(p) = 1/[p(1 - p)]$  and  $\text{Var}(\bar{X}_n) = p(1 - p)/n$ , so  $\bar{X}_n$  is efficient.

## 3 Confidence Intervals and Regions

A point estimate says nothing about its own precision. We therefore attach to it a random set that contains the true parameter with a prescribed probability.

### 3.1 Definition

In the frequentist view  $\theta$  is a fixed number, so a statement such as  $\mathbb{P}(\theta \in [1, 2]) \geq 0.95$  with fixed bounds is meaningless. The randomness must sit in the bounds. For  $\alpha \in (0, 1)$ , a **confidence interval** at level  $1 - \alpha$  is  $[h_1(X), h_2(X)]$  with

$$\inf_{\theta \in \Theta} \mathbb{P}_\theta \big( h_1(X) \leq \theta \leq h_2(X) \big) \geq 1 - \alpha$$

equivalently, the coverage is at least  $1 - \alpha$  for *every*  $\theta$ . For  $\theta \in \mathbb{R}^k$  a **confidence region**  $C(X)$  satisfies  $\mathbb{P}_\theta(\theta \in C(X)) \geq 1 - \alpha$  for all  $\theta$ . The bounds must be functions of the data only, never of the unknown parameter.

#### 3.2 Method 1: Probability Inequalities

**Chebyshev (Bienaymé Tchebychev):** For  $Y$  with finite variance and any  $\epsilon > 0$ ,

$$\mathbb{P}(|Y - \mathbb{E}Y| \geq \epsilon) \leq \frac{\text{Var}(Y)}{\epsilon^2}$$

If  $\hat{\theta}_n$  is unbiased and  $\text{Var}_\theta(\hat{\theta}_n) \leq a$  for all  $\theta$ , then choosing  $\epsilon = \sqrt{a/\alpha}$  gives the interval  $\hat{\theta}_n \pm \sqrt{a/\alpha}$  at level  $1 - \alpha$ , with no distributional assumption.

**Bernoulli Case:**  $\text{Var}_p(\bar{X}_n) = p(1 - p)/n \leq 1/(4n)$ , since  $x(1 - x) \leq 1/4$  on  $(0, 1)$ .

Hence

$$\bigg[ \bar{X}_n - \frac{1}{\sqrt{4n\alpha}}, \quad \bar{X}_n + \frac{1}{\sqrt{4n\alpha}} \bigg]$$

For  $n = 50$  and  $\alpha = 10\%$  the half width is  $1/\sqrt{20} \approx 0.22$ , which is wide for a probability. The price of making no assumptions is conservative intervals.

#### 3.3 Method 2: Pivotal Quantities

A **pivot** is a function of the data and  $\theta$  whose distribution is fully known. Inverting a probability statement about the pivot yields an interval.

**Gaussian Mean, Known Variance:** For  $X_i \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$  with  $\sigma^2$  known,  $\sqrt{n}(\bar{X}_n - \mu)/\sigma \sim \mathcal{N}(0, 1)$ , since a sum of independent normals is normal. Taking symmetric quantiles,

$$\bigg[ \bar{X}_n - \frac{\sigma}{\sqrt{n}} \, q_{1-\alpha/2}, \quad \bar{X}_n + \frac{\sigma}{\sqrt{n}} \, q_{1-\alpha/2} \bigg]$$

If  $\sigma$  were unknown this would not be a confidence interval, because its bounds would depend on an unknown quantity.

**Gaussian Mean, Unknown Variance:** With  $S_n^2 = \frac{1}{n-1} \sum (X_i - \bar{X}_n)^2$ , the facts that  $\bar{X}_n$  and  $S_n^2$  are independent and  $(n - 1)S_n^2/\sigma^2 \sim \chi_{n-1}^2$  give  $\sqrt{n}(\bar{X}_n - \mu)/S_n \sim t_{n-1}$ , and

$$\bar{X}_n \pm t_{n-1}(1 - \alpha/2) \, \frac{S_n}{\sqrt{n}}$$

**Gaussian Variance:** With  $c_{n-1}(\cdot)$  the  $\chi_{n-1}^2$  quantiles,

$$\bigg[ \frac{(n - 1)S_n^2}{c_{n-1}(1 - \alpha/2)}, \quad \frac{(n - 1)S_n^2}{c_{n-1}(\alpha/2)} \bigg]$$

This interval is not symmetric around  $S_n^2$  because the  $\chi^2$  law is skewed.

#### 3.4 Method 3: Asymptotic Intervals

An **asymptotic** confidence interval satisfies  $\lim_{n \rightarrow \infty} \mathbb{P}_\theta(\theta \in [h_1, h_2]) = 1 - \alpha$  (or at least  $\geq$ ) for every  $\theta$ . The recipe is: apply the CLT, replace any unknown scale by a consistent estimator, and justify the replacement with Slutsky’s theorem.

**Bernoulli Proportion:** The CLT gives  $\sqrt{n}(\hat{p} - p)/\sqrt{p(1 - p)} \xrightarrow{d} \mathcal{N}(0, 1)$ , but  $p(1 - p)$  is unknown. Since  $\hat{p} \xrightarrow{\mathbb{P}} p$  and  $x \mapsto \sqrt{x(1 - x)}$  is continuous,  $\sqrt{\hat{p}(1 - \hat{p})} \xrightarrow{\mathbb{P}} \sqrt{p(1 - p)}$ , and Slutsky yields

$$\hat{p} \pm q_{1-\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

**Mean of an Arbitrary Distribution:** For iid data with finite variance only,  $S_n \xrightarrow{\mathbb{P}} \sigma$  and

$$\sqrt{n} \, \frac{\bar{X}_n - \mu}{S_n} \xrightarrow{d} \mathcal{N}(0, 1) \quad \Rightarrow \quad \bar{X}_n \pm q_{1-\alpha/2} \, \frac{S_n}{\sqrt{n}}$$

This is the natural interval for the mean of daily log returns when we are unwilling to assume normality.

**Variance of an Arbitrary Distribution:** With a finite fourth moment, the asymptotic normality of  $V_n$  and a consistent estimator  $\hat{\kappa}$  of the kurtosis give

$$V_n \pm q_{1-\alpha/2} \, V_n \sqrt{\frac{\hat{\kappa} - 1}{n}}$$

**Via the Delta Method:** If  $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} \mathcal{N}(0, s^2(\theta))$  with  $s$  continuous, then  $\hat{\theta}_n \pm q_{1-\alpha/2} \, s(\hat{\theta}_n)/\sqrt{n}$  is asymptotically valid, and the same logic applies to  $g(\hat{\theta}_n)$  with  $s$  replaced by  $|g'(\hat{\theta}_n)| \, s(\hat{\theta}_n)$ .

**Empirical CDF:** For iid data with CDF  $F$ ,  $\hat{F}_n(x) = \frac{1}{n} \sum \mathbb{I}\{X_i \leq x\}$  satisfies  $n\hat{F}_n(x) \sim \text{Binomial}(n, F(x))$ , so  $\hat{F}_n(x)$  is unbiased and consistent for  $F(x)$ , and the Bernoulli interval above applies pointwise.

## 4 Hypothesis Testing

### 4.1 Framework

A test chooses between two mutually exclusive statements about the parameter: the **null**  $H_0 : \theta \in \Theta_0$  and the **alternative**  $H_1 : \theta \in \Theta_1$ , with  $\Theta_0 \cap \Theta_1 = \emptyset$ . A hypothesis is always a claim about  $\theta$ , never about the data (“ $\bar{X}_n = 0$ ” is not a hypothesis). It is **simple** if it pins down the distribution completely (e.g.  $\mu = 0$  with known variance) and **composite** otherwise (e.g.  $\mu \neq 0$ ).

**Decision Rule:** A map  $T : \mathcal{X} \rightarrow \{0, 1\}$ , where  $T = 1$  means reject  $H_0$ . It is usually built from a **test statistic**, a function of the data alone. The **rejection (critical) region** is  $R = \{x : T(x) = 1\}$  and the acceptance region is its complement.

#### 4.2 Errors, Size and Power

|              | $H_0$ <b>true</b> | $H_1$ <b>true</b> |
|--------------|-------------------|-------------------|
| Reject $H_0$ | Type I error      | Correct           |
| Accept $H_0$ | Correct           | Type II error     |

- Type I risk:**  $\alpha(\theta) = \mathbb{P}_\theta(T = 1)$  for  $\theta \in \Theta_0$
- Size:**  $\alpha^* = \sup_{\theta \in \Theta_0} \mathbb{P}_\theta(T = 1)$ ; the test has **level**  $\alpha$  if  $\alpha^* \leq \alpha$
- Type II risk:**  $\beta(\theta) = \mathbb{P}_\theta(T = 0)$  for  $\theta \in \Theta_1$
- Power function:**  $\pi(\theta) = \mathbb{P}_\theta(T = 1)$  on all of  $\Theta$ , so  $\pi = \alpha$  on  $\Theta_0$  and  $\pi = 1 - \beta$  on  $\Theta_1$

Ideally  $\pi$  would be 0 on  $\Theta_0$  and 1 on  $\Theta_1$ , but when the two sets share a boundary and  $\pi$  is continuous this is impossible. Following Neyman, we treat the null asymmetrically: we cap the Type I risk at a small  $\alpha$  and then seek high power.

#### 4.3 Gaussian Mean Tests

Let  $X_i \stackrel{\text{iid}}{\sim} \mathcal{N}(\theta, \sigma^2)$  with  $\sigma$  known, so  $\bar{X}_n \sim \mathcal{N}(\theta, \sigma^2/n)$ .

**One Sided,  $H_0 : \theta \leq \theta_0$  vs  $H_1 : \theta > \theta_0$ :** Reject when  $\bar{X}_n > c$ . The size condition  $\sup_{\theta \leq \theta_0} \mathbb{P}_\theta(\bar{X}_n > c) \leq \alpha$  is binding at  $\theta = \theta_0$  because  $\Phi$  is increasing, giving

$$c = \theta_0 + q_{1-\alpha} \, \frac{\sigma}{\sqrt{n}}, \qquad \pi(\theta) = \Phi \left( \frac{\sqrt{n}(\theta - \theta_0)}{\sigma} - q_{1-\alpha} \right)$$

The power increases in  $\theta$ , equals  $\alpha$  at the boundary, and becomes steeper as  $n$  grows.

**Two Sided,  $H_0 : \theta = \theta_0$  vs  $H_1 : \theta \neq \theta_0$ :** Reject when  $|\bar{X}_n - \theta_0| > q_{1-\alpha/2} \, \sigma/\sqrt{n}$ . Its power is

$$\pi(\theta) = 1 - \Phi(q_{1-\alpha/2} - \delta) + \Phi(-q_{1-\alpha/2} - \delta), \qquad \delta = \frac{\sqrt{n}(\theta - \theta_0)}{\sigma}$$

**Unknown Variance:** Replace  $\sigma$  by  $S_n$  and normal quantiles by  $t_{n-1}$  quantiles (the one sample  $t$  test).

#### 4.4 Duality with Confidence Intervals

**Proposition:** If  $C(X)$  is a confidence interval at level  $1 - \alpha$ , then the test that rejects  $H_0$  whenever  $C(X) \cap \Theta_0 = \emptyset$  has level  $\alpha$ . Indeed, for  $\theta \in \Theta_0$ , rejection forces  $\theta \notin C(X)$ , which has probability at most  $\alpha$ .

**Direction Matters:** For the one sided problem above, the lower bounded interval  $[\bar{X}_n - q_{1-\alpha}\sigma/\sqrt{n}, \infty)$  reproduces the test. The upper bounded interval

$(-\infty, \bar{X}_n + q_{1-\alpha}\sigma/\sqrt{n}]$  always meets  $\Theta_0 = (-\infty, \theta_0]$ , so the resulting test never rejects: valid, but useless. For a one sided test, start from the interval pointing *away* from  $\Theta_0$ .

#### 4.5 Asymptotic Tests

A sequence of tests  $(T_n)$  has **asymptotic level**  $\alpha$  if  $\limsup_n \mathbb{P}_\theta(T_n = 1) \leq \alpha$  for all  $\theta \in \Theta_0$ . Asymptotic confidence intervals give asymptotic tests in the same way as above.

**Testing a Proportion:** For  $H_0 : p = p_0$  vs  $H_1 : p \neq p_0$ , under  $H_0$  the CLT gives  $\sqrt{n}(\hat{p} - p_0)/\sqrt{p_0(1 - p_0)} \xrightarrow{d} \mathcal{N}(0, 1)$ , so we reject when

$$|\hat{p} - p_0| > q_{1-\alpha/2} \sqrt{\frac{p_0(1 - p_0)}{n}}$$

With  $p_0 = 0.1$ ,  $n = 400$  and  $\alpha = 5\%$  the threshold is  $1.96 \times 0.015 \approx 0.029$ .

#### 4.6 p-values

Rather than fixing  $\alpha$  in advance, we can ask for which levels the observed data lead to rejection. Assuming the rejection regions are nested ( $\alpha_1 \leq \alpha_2 \Leftrightarrow R_{\alpha_1} \subseteq R_{\alpha_2}$ ), the **p-value** is

$$\hat{\alpha}(x) = \inf\{\alpha \in [0, 1] : x \in R_\alpha\}$$

the smallest level at which  $H_0$  is rejected. A small p-value is evidence against  $H_0$ ; we reject at level  $\alpha$  exactly when  $\hat{\alpha}(x) \leq \alpha$  (up to the boundary).

**Tail Probability Form:** If the test rejects when  $h(X) > c_\alpha$ , then

$$\hat{\alpha}(x) = \sup_{\theta \in \Theta_0} \mathbb{P}_\theta(h(X) > h(x))$$

the null probability of obtaining a statistic as extreme as, or more extreme than, the value we actually saw. For the one sided Gaussian test with  $\sigma = 1$  and  $\theta_0 = 0$ ,  $\hat{\alpha} = 1 - \Phi(\sqrt{n}\bar{x}_n)$ ; for instance  $n = 64$  and  $\bar{x}_n = 0.25$  give  $1 - \Phi(2) \approx 0.023$ , a rejection at 5% but not at 1%.

#### 4.7 Likelihood Ratio Tests

**Neyman Pearson Lemma:** For two simple hypotheses  $H_0 : \theta = \theta_0$  vs  $H_1 : \theta = \theta_1$ , the test rejecting when

$$\Lambda(x) = \frac{L(x; \theta_1)}{L(x; \theta_0)} > k$$

with  $k$  chosen so that its size is  $\alpha$ , is the most powerful test of level  $\alpha$ . Intuitively, we reject when the data are much more likely under the alternative than under the null.

**Generalised LRT:** For composite hypotheses,  $\lambda(x) = \sup_{\Theta_0} L/\sup_{\Theta} L$ , and under regularity  $-2\log \lambda(X) \xrightarrow{d} \chi_r^2$  under  $H_0$ , where  $r$  is the number of restrictions imposed by  $H_0$ .

## 5 Linear Regression

#### 5.1 The Model

A regression model explains a **response**  $Y_i$  through **explanatory** vectors  $x_i \in \mathbb{R}^p$  (observed, treated as fixed), a regression function  $g$  and centred, uncorrelated errors:  $Y_i = g(x_i) + \varepsilon_i$ . When  $g$  is linear in the unknown coefficients we have the **linear model**

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbf{Y}, \boldsymbol{\varepsilon} \in \mathbb{R}^n, \; \mathbf{X} \in \mathbb{R}^{n \times p}, \; \boldsymbol{\beta} \in \mathbb{R}^p$$

Simple linear regression  $Y_i = a + bx_i + \varepsilon_i$  has rows  $(1, x_i)$ ; polynomial regression  $Y_i = a_0 + a_1x_i + \cdots + a_mx_i^m + \varepsilon_i$  is still linear in  $\boldsymbol{\beta}$ , with  $\mathbf{X}$  of size  $n \times (m + 1)$ . A CAPM regression of excess stock returns on excess market returns is a simple linear regression with  $\boldsymbol{\beta} = (\alpha, \beta)^T$ .

**Assumptions:**

- (A1)**  $\text{rank}(\mathbf{X}) = p$ , equivalently  $\mathbf{X}^T\mathbf{X}$  is invertible (this requires  $n \geq p$ )
- (A2)**  $\mathbb{E}(\boldsymbol{\varepsilon}) = 0$  and  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 I_n$ : centred, homoscedastic, uncorrelated errors
- (A3)**  $\boldsymbol{\varepsilon} \sim \mathcal{N}_n(0, \sigma^2 I_n)$ , which implies (A2) and makes the errors independent

Uncorrelated is weaker than independent; (A3) upgrades one to the other because the vector is jointly Gaussian.

#### 5.2 Least Squares

The **least squares estimator** minimises  $\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \sum_{i=1}^n (Y_i - x_i^T\boldsymbol{\beta})^2$ . The matrix  $\mathbf{X}^T\mathbf{X}$  is always symmetric positive semi-definite, and positive definite exactly under (A1).

**Theorem:** Under (A1) the minimiser exists, is unique and equals

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y}$$

**Geometric View:** Let  $D = \text{Im}(\mathbf{X}) = \{\mathbf{X}\boldsymbol{\beta} : \boldsymbol{\beta} \in \mathbb{R}^p\} \subseteq \mathbb{R}^n$ , the span of the columns of  $\mathbf{X}$ , of dimension  $p$ . Minimising the distance from  $\mathbf{Y}$  to  $D$  means projecting  $\mathbf{Y}$  orthogonally onto  $D$ . Orthogonality of the residual to every column gives the **normal equations**  $\mathbf{X}^T(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = 0$ , hence the formula. The projection matrix onto  $D$  is the **hat matrix**

$$H = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T, \quad \hat{\mathbf{Y}} = H\mathbf{Y}$$

$H$  is symmetric and idempotent ( $H^2 = H$ ),  $HX = X$ , and  $I_n - H$  projects onto the orthogonal complement of  $D$ .

**Moments:** Under (A1) and (A2),

$$\mathbb{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}, \quad \text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}$$

since  $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\varepsilon}$  and  $\text{Cov}(A\boldsymbol{\varepsilon}) = A\text{Cov}(\boldsymbol{\varepsilon})A^T$ .

**Gauss Markov Theorem:** A linear estimator  $\hat{\boldsymbol{\beta}} = L\mathbf{Y}$  is unbiased for all  $\boldsymbol{\beta}$  if and only if  $L\mathbf{X} = I_p$ . Among all such estimators,  $\text{Cov}(\hat{\boldsymbol{\beta}}) - \text{Cov}(\hat{\boldsymbol{\beta}})$  is positive semi-definite, so OLS is the **best linear unbiased estimator** (BLUE).

#### 5.3 Residuals and Error Variance

The residual vector is  $\hat{\boldsymbol{\varepsilon}} = \mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}} = (I_n - H)\boldsymbol{\varepsilon}$  and the **residual sum of squares** is  $RSS = \|\hat{\boldsymbol{\varepsilon}}\|^2 = \boldsymbol{\varepsilon}^T(I_n - H)\boldsymbol{\varepsilon}$ .

**Unbiased Variance Estimator:** Using  $\mathbb{E}[\boldsymbol{\varepsilon}^T A\boldsymbol{\varepsilon}] = \text{tr}(A\text{Cov}(\boldsymbol{\varepsilon}))$  for centred  $\boldsymbol{\varepsilon}$ , and  $\text{tr}(I_n - H) = \text{rank}(I_n - H) = n - p$ ,

$$\mathbb{E}(RSS) = (n - p)\sigma^2 \;\Rightarrow\; \hat{\sigma}^2 = \frac{RSS}{n - p}$$

is unbiased for  $\sigma^2$ , and  $\widehat{\text{Cov}}(\hat{\boldsymbol{\beta}}) = \hat{\sigma}^2(\mathbf{X}^T\mathbf{X})^{-1}$ . We need  $p < n$ ; with  $p = n$  the fit is exact,  $\hat{\mathbf{Y}} = \mathbf{Y}$ , and nothing is left to estimate the noise.

**Goodness of Fit:** With  $TSS = \sum (Y_i - \bar{Y})^2$  and an intercept in the model,  $R^2 = 1 - RSS/TSS$ .

#### 5.4 Gaussian Linear Model

Under (A1) and (A3), each  $Y_i \sim \mathcal{N}(x_i^T\boldsymbol{\beta}, \sigma^2)$  independently (not identically distributed, since the means differ). Maximising the Gaussian likelihood in  $\boldsymbol{\beta}$  is the same as minimising the RSS, so **OLS coincides with the MLE** of  $\boldsymbol{\beta}$  (the MLE of  $\sigma^2$  is  $RSS/n$ , which is biased).

**Distributional Results:**

- $\hat{\boldsymbol{\beta}} \sim \mathcal{N}_p(\boldsymbol{\beta}, \sigma^2(\mathbf{X}^T\mathbf{X})^{-1})$
- $\hat{\boldsymbol{\beta}}$  and  $\hat{\sigma}^2$  are independent
- $(n - p)\hat{\sigma}^2/\sigma^2 \sim \chi_{n-p}^2$

Parts 2 and 3 follow from Cochran’s theorem, since  $H$  and  $I_n - H$  are orthogonal projections.

**Pivots:** Writing  $v_{jj} = [(\mathbf{X}^T\mathbf{X})^{-1}]_{jj}$  and  $\hat{\sigma}_{\hat{\beta}_j} = \hat{\sigma}\sqrt{v_{jj}}$ ,

$$T_j = \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}_{\hat{\beta}_j}} \sim t_{n-p}, \quad F = \frac{(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})^T(\mathbf{X}^T\mathbf{X})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})}{p\hat{\sigma}^2} \sim F_{p, n-p}$$

**Confidence Sets at Level  $1 - \alpha$ :**

- For  $\beta_j$ :  $\hat{\beta}_j \pm t_{n-p}(1 - \alpha/2) \hat{\sigma}\sqrt{v_{jj}}$
- For  $\sigma^2$ :  $\left[ \frac{(n-p)\hat{\sigma}^2}{c_{n-p}(1-\alpha/2)}, \; \frac{(n-p)\hat{\sigma}^2}{c_{n-p}(\alpha/2)} \right]$
- For  $\boldsymbol{\beta}$ : the ellipsoid  $\{\boldsymbol{\beta} : F \leq f_{p, n-p}(1 - \alpha)\}$

When  $\sigma^2$  is known, the first result gives a  $\chi_p^2$  ellipsoid directly.

**Significance Test:** For  $H_0 : \beta_k = 0$  vs  $H_1 : \beta_k \neq 0$ , reject at level  $\alpha$  when  $0 \notin \hat{\beta}_k \pm t_{n-p}(1 - \alpha/2)\hat{\sigma}\sqrt{v_{kk}}$ , i.e. when  $|T_k| > t_{n-p}(1 - \alpha/2)$  with  $\beta_k = 0$ . A rejected null means  $\beta_k$  is **significant**. This is the  $t$  value reported by regression software.

#### 5.5 Prediction

For a new design point  $x_{n+1}$  with  $Y_{n+1} = x_{n+1}^T\boldsymbol{\beta} + \varepsilon_{n+1}$  and  $\varepsilon_{n+1}$  independent of the sample, the natural prediction is  $\hat{Y}_{n+1} = x_{n+1}^T\hat{\boldsymbol{\beta}}$ . The prediction error combines estimation error and fresh noise:

$$Y_{n+1} - \hat{Y}_{n+1} = x_{n+1}^T(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}) + \varepsilon_{n+1} \sim \mathcal{N}\Big(0, \; \sigma^2(1 + x_{n+1}^T(\mathbf{X}^T\mathbf{X})^{-1}x_{n+1})\Big)$$

The two terms are independent, and only the first shrinks as  $n$  grows. The **prediction interval** at level  $1 - \alpha$  is

$$x_{n+1}^T\hat{\boldsymbol{\beta}} \pm t_{n-p}(1 - \alpha/2) \hat{\sigma}\sqrt{1 + x_{n+1}^T(\mathbf{X}^T\mathbf{X})^{-1}x_{n+1}}$$

Dropping the 1 under the root gives the narrower confidence interval for the mean response  $x_{n+1}^T\boldsymbol{\beta}$ .

#### 5.6 Model Selection

With  $k$  estimated parameters and maximised likelihood  $\hat{L}$ ,

$$\text{AIC} = 2k - 2\log \hat{L}, \quad \text{BIC} = k\log n - 2\log \hat{L}$$

Smaller is better. BIC penalises extra parameters more heavily once  $n \geq 8$ , so it tends to choose smaller models.

## 6 Bayesian Inference

#### 6.1 Change of Viewpoint

In the frequentist approach  $\theta$  is a fixed but unknown point of  $\Theta$ . In the Bayesian approach  $\theta$  is itself a random variable, given a **prior** distribution that encodes what we believe before seeing data. Observations are modelled *conditionally* on  $\theta$ , and independence statements are understood conditionally on  $\theta$ .

#### 6.2 Probability Review

**Conditional Probability:**  $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$  for  $\mathbb{P}(B) > 0$ .

**Total Probability:** For a partition  $(A_k)$ ,  $\mathbb{P}(B) = \sum_k \mathbb{P}(B \mid A_k)\mathbb{P}(A_k)$ .

**Bayes’ Theorem:**

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B \mid A)\mathbb{P}(A) + \mathbb{P}(B \mid A^c)\mathbb{P}(A^c)}$$

**Conditional Independence:**  $A$  and  $B$  are independent given  $C$  if  $\mathbb{P}(A \cap B \mid C) = \mathbb{P}(A \mid C)\mathbb{P}(B \mid C)$ .

**Densities:**  $f(x \mid y) = f(x, y)/\int f(x, y) \, dx$ , and the continuous form of Bayes’ theorem is

$$f(y \mid x) = \frac{f(x \mid y) \, g(y)}{\int f(x \mid y) \, g(y) \, dy}$$

#### 6.3 Prior and Posterior

A **Bayesian model** consists of a prior  $\pi$  on  $\Theta$  and the conditional law  $f(x \mid \theta)$  of  $X = (X_1, \dots, X_n)$  given  $\theta$ . The object of interest is the **posterior**, the law of  $\theta$  given the data. Whether prior and data are discrete or continuous, Bayes’ theorem gives (with sums replacing integrals where needed)

$$\pi(\theta \mid x) = \frac{f(x \mid \theta) \, \pi(\theta)}{\int_{\Theta} f(x \mid u) \, \pi(u) \, du} \;\propto\; f(x \mid \theta) \, \pi(\theta)$$

The denominator is a normalising constant. In practice we compute the numerator as a function of  $\theta$ , recognise the kernel of a known family, and read off the constant. The prior is often chosen partly for tractability, and its choice is the most debated part of the approach.

### 6.4 Conjugate Families

A family  $\mathcal{F}$  of priors is **conjugate** for  $f(x|\theta)$  if the posterior stays in  $\mathcal{F}$  whenever the prior does; only the parameters are updated. For  $n$  conditionally iid observations with  $s = \sum x_i$ :

| Likelihood                                         | Prior                      | Posterior                                                               |
|----------------------------------------------------|----------------------------|-------------------------------------------------------------------------|
| Bernoulli( $\theta$ )                              | Beta( $a, b$ )             | Beta( $a + s, b + n - s$ )                                              |
| Poisson( $\lambda$ )                               | Gamma( $\alpha, \beta$ )   | Gamma( $\alpha + s, \beta + n$ )                                        |
| Exp( $\lambda$ )                                   | Gamma( $\alpha, \beta$ )   | Gamma( $\alpha + n, \beta + s$ )                                        |
| $\mathcal{N}(\theta, \sigma^2)$ , $\sigma^2$ known | $\mathcal{N}(\mu, \tau^2)$ | $\mathcal{N}(m_n, v_n)$ , see below                                     |
| $\mathcal{N}(\mu, 1/\theta)$ , $\mu$ known         | Gamma( $\alpha, \beta$ )   | Gamma( $\alpha + \frac{1}{2}, \beta + \frac{1}{2} \sum (x_i - \mu)^2$ ) |

For the normal mean, precisions add and the posterior mean is a precision weighted average:

$$\frac{1}{v_n} = \frac{1}{\tau^2} + \frac{n}{\sigma^2}, \quad m_n = v_n \left( \frac{\mu}{\tau^2} + \frac{n\bar{x}}{\sigma^2} \right)$$

With one observation this reads  $m_1 = \frac{\sigma^2\mu + \tau^2x}{\sigma^2 + \tau^2} = x - \frac{\sigma^2}{\sigma^2 + \tau^2}(x - \mu)$  and  $v_1 = \frac{\sigma^2\tau^2}{\sigma^2 + \tau^2}$ . As  $n$  grows the data dominate and  $m_n \rightarrow \bar{x}$ .

**Deriving a Posterior (Beta Bernoulli):**

$$\pi(\theta|x) \propto \theta^s(1-\theta)^{n-s} \cdot \theta^{a-1}(1-\theta)^{b-1} = \theta^{a+s-1}(1-\theta)^{b+n-s-1}$$

which is the kernel of Beta( $a + s, b + n - s$ ), with posterior mean  $\frac{a+s}{a+b+n}$ .

### 6.5 Risk and Bayes Estimators

**Integrated Risk:** The frequentist risk  $R(\theta, T)$  depends on  $\theta$ , so estimators are hard to rank. Averaging over the prior gives a single number,

$$r(\pi, T) = \mathbb{E}^\pi[R(\theta, T)]$$

and any two estimators become comparable.

**Bayes Estimator:**  $T^\pi$  minimising  $r(\pi, T)$ ; its value  $r(\pi) = r(\pi, T^\pi)$  is the **Bayes risk**.

**Posterior Risk:**  $\rho(\pi, T|x) = \int_{\Theta} L(\theta, T(x)) \pi(\theta|x) d\theta$ . The integrated risk is its average over the marginal law  $m$  of the data,  $r(\pi, T) = \int \rho(\pi, T(x)|x) dm(x)$ . Hence, when  $r$  is finite, we find the Bayes estimator by minimising the posterior risk separately for each observed  $x$ .

**Standard Results:**

- Quadratic loss  $(\theta - d)^2$ : the Bayes estimator is the **posterior mean**  $\mathbb{E}(\theta|X)$
- Absolute loss  $|\theta - d|$ : the Bayes estimator is the **posterior median**

For the quadratic case, expanding  $\int (T - \theta)^2 \pi(\theta|x) d\theta = T^2 - 2T \mathbb{E}(\theta|x) + \mathbb{E}(\theta^2|x)$  gives a quadratic in  $T$  minimised at  $T = \mathbb{E}(\theta|x)$ .

**Admissibility:** A Bayes estimator is admissible if it is unique, or if its Bayes risk is finite.

**MAP Estimator:**  $\hat{\theta}_{MAP} = \arg \max_{\theta} \pi(\theta|x) = \arg \max_{\theta} \pi(\theta) L(x; \theta)$ . It is the MLE reweighted by the prior; with a flat prior the two coincide.

### 6.6 Non Informative Priors

**Laplace:** a uniform prior on  $\Theta$ . It may be **improper** (infinite mass), which is acceptable if the posterior is proper, but it is not invariant under reparameterisation (flat in  $\theta$  is not flat in  $\theta^2$ ).

**Jeffreys:**  $\pi(\theta) \propto \sqrt{\det I(\theta)}$ , with  $I$  the Fisher information. It is invariant under reparameterisation. For Bernoulli,  $I(\theta) = 1/[\theta(1-\theta)]$  gives Beta( $\frac{1}{2}, \frac{1}{2}$ ).

### 6.7 Credible Intervals

A **credible region** at level  $1 - \alpha$  is a set  $C(x)$  with  $\pi(\theta \in C(x)|x) = 1 - \alpha$ . Unlike a confidence interval, this is a direct probability statement about  $\theta$  given the observed data.

**Equal Tailed Example:** If the posterior is  $\mathcal{N}(0.5, 0.25)$ , the symmetric 95% credible interval is  $0.5 \pm 1.96 \times 0.5 = [-0.48, 1.48]$ .

**HPD Region:** The highest posterior density region  $\{\theta : \pi(\theta|x) \geq k\}$  is the shortest region with the required mass; for a symmetric unimodal posterior it coincides with the equal tailed interval.

### 6.8 Bayesian Testing

For  $H_0 : \theta \in \Theta_0$  vs  $H_1 : \theta \in \Theta_1$ , take the weighted 0-1 loss that costs  $a_0 > 0$  for rejecting a true  $H_0$  and  $a_1 > 0$  for accepting a false one. The posterior risk of a binary rule is  $a_0 \mathbb{P}\{T = 1\} \pi(\Theta_0|x) + a_1 \mathbb{P}\{T = 0\} \pi(\Theta_1|x)$ , which is minimised by reject  $H_0 \iff a_1 \pi(\Theta_1|x) > a_0 \pi(\Theta_0|x)$

This is a posterior analogue of the likelihood ratio test: reject when the posterior odds of  $H_1$  exceed the ratio of costs  $a_0/a_1$ .

## 7 Time Series and ARMA Models

### 7.1 Basic Notions

By a **time series** we mean a random process observed at discrete dates, that is, a family  $(X_t)_{t \in \mathbb{Z}}$  (or  $t \in \mathbb{N}$ ) of random variables. For a fixed outcome  $\omega$ , the path  $t \mapsto X_t(\omega)$  is a **trajectory**; in practice we observe a single trajectory and treat it as one realisation of the process. This contrasts with continuous time models such as Brownian motion, where  $t$  ranges over an interval. We assume throughout that the process is **square integrable**: each  $X_t$  has finite mean and variance.

**White Noise:**  $(\varepsilon_t)$  is a WN if  $\mathbb{E}(\varepsilon_t)$  and  $\text{Var}(\varepsilon_t)$  do not depend on  $t$  and  $\text{Cov}(\varepsilon_t, \varepsilon_{t+h}) = 0$  for all  $h \neq 0$ . It is **centred** if the mean is zero; we write  $\text{WN}(0, \sigma^2)$ . A **strong** WN is additionally iid.

**(Weak) Stationarity:**  $\mathbb{E}(X_t)$  and  $\text{Var}(X_t)$  do not depend on  $t$ , and  $\text{Cov}(X_t, X_{t+h})$  depends only on the lag  $h$ . Every WN is stationary, but not conversely.

**Autocovariance and Autocorrelation:** For a stationary process

$$\gamma(h) = \text{Cov}(X_t, X_{t+h}), \quad \rho(h) = \frac{\gamma(h)}{\gamma(0)}$$

with  $\gamma(0) = \text{Var}(X_t)$ ,  $\rho(0) = 1$ , and  $\gamma(-h) = \gamma(h)$ , so it suffices to report  $h \geq 1$ . The ACF measures how strongly the series depends on its own past, which is what makes prediction possible.

**Sample Versions:**

$$\hat{\gamma}_n(h) = \frac{1}{n} \sum_{t=1}^{n-h} (X_t - \bar{X}_n)(X_{t+h} - \bar{X}_n), \quad \hat{\rho}_n(h) = \frac{\hat{\gamma}_n(h)}{\hat{\gamma}_n(0)}$$

(some texts divide by  $n - h$  instead of  $n$ ).

**Sum of Uncorrelated Stationary Processes:** If  $(X_t)$  and  $(Y_t)$  are stationary with  $\text{Cov}(X_t, Y_s) = 0$  for all  $s, t$ , then  $X_t + Y_t$  is stationary with  $\gamma_{X+Y} = \gamma_X + \gamma_Y$ , by bilinearity of the covariance.

### 7.2 Testing for White Noise

Let  $(\varepsilon_t)$  be an iid WN with finite fourth moment. Then for each fixed  $h \geq 1$ ,  $\sqrt{n} \hat{\rho}_n(h) \xrightarrow{d} \mathcal{N}(0, 1)$ , and  $\hat{\rho}_n(1), \hat{\rho}_n(2), \dots$  are asymptotically independent. This justifies the  $\pm 1.96/\sqrt{n}$  bands drawn on sample ACF plots.

**Portmanteau Tests:** For  $H_0 : \rho(1) = \dots = \rho(h) = 0$  against the alternative that at least one is nonzero,

$$Q_n^{BP}(h) = n \sum_{j=1}^h \hat{\rho}_n^2(j), \quad Q_n^{LB}(h) = n(n+2) \sum_{j=1}^h \frac{\hat{\rho}_n^2(j)}{n-j}$$

both converge to  $\chi_h^2$  under  $H_0$  (Box Pierce and Ljung Box respectively). We reject when  $Q > \chi_h^2(1 - \alpha)$ . Ljung Box has better small sample behaviour. When applied to residuals of a fitted ARMA( $p, q$ ), the degrees of freedom are reduced to  $h - p - q$ .

### 7.3 Lag and Difference Operators

**Lag (Backshift) Operator:**  $LX_t = X_{t-1}$ , so  $L^k X_t = X_{t-k}$  and, by convention,  $L^{-k} X_t = X_{t+k}$ . Applied to a constant series it returns the same constant. Some texts write  $B$  instead of  $L$ .

**Lag Polynomials:**

$$\phi(L) = 1 - \sum_{i=1}^p \phi_i L^i, \quad \theta(L) = 1 + \sum_{i=1}^q \theta_i L^i$$

**Difference Operators:**  $\Delta X_t = (1 - L)X_t = X_t - X_{t-1}$  and  $\Delta^k = (1 - L)^k$ . The **seasonal difference** of period  $d$  is  $\Delta_d X_t = (1 - L^d)X_t = X_t - X_{t-d}$ . Note that

$\Delta^k \neq \Delta_k$ .

**Removing Trends:** If  $Y_t = c_0 + c_1 t + X_t$  then  $\Delta Y_t = c_1 + \Delta X_t$ , so a linear trend becomes a constant. In general  $\Delta^k$  reduces a polynomial trend of degree  $k$  to a constant.

**Removing Seasonality:** If  $X_t = s_t + Y_t$  with  $s_{t+d} = s_t$ , then  $\Delta_d X_t = Y_t - Y_{t-d}$  and the seasonal part disappears.

**Multiplicative Components:** If  $X_t = m_t s_t Y_t$  with all terms positive, take logarithms first to make the decomposition additive.

**Box Jenkins Workflow:** Identify and remove the seasonal component with  $\Delta_d$ , remove the trend with  $\Delta$  (as many times as needed), check that what remains looks stationary, then fit a stationary model such as ARMA and validate its residuals.

**Random Walk:**  $X_t = X_{t-1} + \varepsilon_t$  with  $X_0 = 0$  gives  $X_t = \sum_{s=1}^t \varepsilon_s$ ,  $\text{Var}(X_t) = t\sigma^2$  and  $\text{Cov}(X_s, X_t) = \sigma^2 \min(s, t)$ . It is *not* stationary, but  $\Delta X_t = \varepsilon_t$  is. Log prices are often modelled this way, which is why we work with returns.

### 7.4 ARMA Models

**Definition:** With  $(\varepsilon_t) \sim \text{WN}(0, \sigma^2)$ ,

$$X_t = \sum_{i=1}^p \phi_i X_{t-i} + \varepsilon_t + \sum_{j=1}^q \theta_j \varepsilon_{t-j} \iff \phi(L)X_t = \theta(L)\varepsilon_t$$

Setting  $q = 0$  gives an **AR**( $p$ ) model,  $\phi(L)X_t = \varepsilon_t$ ; setting  $p = 0$  gives an **MA**( $q$ ) model,  $X_t = \theta(L)\varepsilon_t$ .

**Causal (Linear) Process:**  $X_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$  with  $\sum |\psi_j| < \infty$ , i.e.  $X_t$  depends only on present and past shocks (an MA( $\infty$ )). Absolute summability makes the series converge almost surely and in  $L^2$ , and allows expectation and summation to be interchanged.

**Invertible Process:**  $\varepsilon_t = \sum_{j=0}^{\infty} \pi_j X_{t-j}$  with  $\sum |\pi_j| < \infty$ , i.e. the shocks can be recovered from present and past observations (an AR( $\infty$ ) representation).

**The Three Theorems:** Suppose  $\phi(z)$  and  $\theta(z)$  have *no common roots*. Then

| Property                   | Condition                               |
|----------------------------|-----------------------------------------|
| Stationary solution exists | $\phi(z) \neq 0$ for all $ z  = 1$      |
| Causal                     | $\phi(z) \neq 0$ for all $ z  \leq 1$   |
| Invertible                 | $\theta(z) \neq 0$ for all $ z  \leq 1$ |

so roots must lie off the unit circle for stationarity and strictly outside the closed unit disc for causality or invertibility. The coefficients come from the power series

$$\psi(z) = \sum_{j \geq 0} \psi_j z^j = \frac{\theta(z)}{\phi(z)}, \quad \pi(z) = \sum_{j \geq 0} \pi_j z^j = \frac{\phi(z)}{\theta(z)}, \quad |z| \leq 1$$

**Checking a Given ARMA in Practice:**

- Write down  $\phi$  and  $\theta$  from the model
- Find all roots of each polynomial
- If they share a root, cancel the common factor (coefficient redundancy) and repeat with the reduced polynomials
- Apply the three conditions above

For instance  $(1 - 0.4L)X_t = (1 - 0.4L)\varepsilon_t$  reduces to  $X_t = \varepsilon_t$ , a white noise.

**Worked Illustration:** Consider  $X_t - 0.5X_{t-1} = \varepsilon_t + 0.4\varepsilon_{t-1}$ . Here  $\phi(z) = 1 - 0.5z$  has root  $z = 2$  and  $\theta(z) = 1 + 0.4z$  has root  $z = -2.5$ ; there is no common root and both lie outside the unit disc, so the unique stationary solution is causal and invertible. Expanding,

$$\psi(z) = (1 + 0.4z) \sum_{j \geq 0} 0.5^j z^j \Rightarrow \psi_0 = 1, \quad \psi_j = 0.9 \times 0.5^{j-1} \quad (j \geq 1)$$

**Moments of a Linear Process:** For causal  $X_t = \sum_j \psi_j \varepsilon_{t-j}$ ,

$$\mathbb{E}(X_t) = 0, \quad \gamma(h) = \sigma^2 \sum_{k=0}^{\infty} \psi_k \psi_{k+h}$$

which do not depend on  $t$ , so a causal ARMA is stationary.

**MA**( $q$ ): With  $\theta_0 = 1$ ,  $\gamma(h) = \sigma^2 \sum_{j=0}^{q-h} \theta_j \theta_{j+h}$  for  $0 \leq h \leq q$  and  $\gamma(h) = 0$  for  $h > q$ . The ACF **cuts off** after lag  $q$ . An MA( $q$ ) is always stationary.

**AR(1):**  $X_t = \phi X_{t-1} + \varepsilon_t$  is causal iff  $|\phi| < 1$ , with  $\psi_j = \phi^j$ ,  $\gamma(h) = \sigma^2 \phi^{|h|} / (1 - \phi^2)$  and  $\rho(h) = \phi^{|h|}$ ; the ACF decays geometrically. If  $|\phi| > 1$  a stationary solution still exists but depends on future shocks;  $\phi = 1$  is the random walk.

### 7.5 Identification and Estimation

**Choosing  $p$  and  $q$ :**

| Model          | ACF                    | PACF                   |
|----------------|------------------------|------------------------|
| AR( $p$ )      | Tails off              | Cuts off after lag $p$ |
| MA( $q$ )      | Cuts off after lag $q$ | Tails off              |
| ARMA( $p, q$ ) | Tails off              | Tails off              |

The partial autocorrelation at lag  $h$  is the correlation between  $X_t$  and  $X_{t-h}$  after removing the linear effect of  $X_{t-1}, \dots, X_{t-h+1}$ . Information criteria (AIC, BIC) compare candidate orders directly.

**Estimating Coefficients:**

- AR( $p$ ):** regress  $X_t$  on  $(X_{t-1}, \dots, X_{t-p})$  by least squares, as in linear regression
- ARMA( $p, q$ ):** a two step regression in the spirit of Durbin (1960): fit a long AR to estimate the unobserved shocks, then regress  $X_t$  on lagged  $X$  and lagged estimated shocks
- Maximum likelihood:** requires a distributional assumption on the noise (typically Gaussian), under which the likelihood can be written through the one step prediction errors

**Useful Identity:**

$\text{Cov}(A + B, C + D) = \text{Cov}(A, C) + \text{Cov}(A, D) + \text{Cov}(B, C) + \text{Cov}(B, D)$ .

## 8 ARCH and GARCH Models

### 8.1 Motivation

Financial returns are close to uncorrelated, yet large moves tend to be followed by large moves of either sign (**volatility clustering**), and their distribution has heavier tails than the normal. ARMA models describe the conditional mean only; ARCH and GARCH models describe a **conditional variance** that changes over time. *Heteroscedasticity* means unequal variability, and *conditional* heteroscedasticity means the variance given the past depends on time. ARCH was introduced by Engle (1982) and generalised to GARCH by Bollerslev (1986).

**Setting:**  $(\varepsilon_t)$  is a centred **strong** white noise with unit variance (iid, mean 0, variance 1), and  $\mathcal{F}_t = \sigma\{X_s, \sigma_s : s \leq t\}$  is the information available at time  $t$ .

**Conditional Variance:**

$\text{Var}(Y \mid \mathcal{G}) = \mathbb{E}[(Y - \mathbb{E}(Y \mid \mathcal{G}))^2 \mid \mathcal{G}] = \mathbb{E}(Y^2 \mid \mathcal{G}) - \mathbb{E}(Y \mid \mathcal{G})^2$ .

### 8.2 ARCH(1)

$$X_t = \sigma_t \varepsilon_t, \quad \sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2, \quad \alpha_0 > 0, \; \alpha_1 \geq 0$$

**Conditional Mean:**  $\sigma_t$  is known at  $t - 1$  and  $\varepsilon_t$  is independent of  $\mathcal{F}_{t-1}$ , so

$$\mathbb{E}(X_t \mid \mathcal{F}_{t-1}) = \sigma_t \mathbb{E}(\varepsilon_t) = 0, \quad \mathbb{E}(X_t) = \mathbb{E}\big[\mathbb{E}(X_t \mid \mathcal{F}_{t-1})\big] = 0$$

by the tower property.

**Conditional Variance:** Since the conditional mean is zero,

$$\text{Var}(X_t \mid \mathcal{F}_{t-1}) = \mathbb{E}(\sigma_t^2 \varepsilon_t^2 \mid \mathcal{F}_{t-1}) = \sigma_t^2 \mathbb{E}(\varepsilon_t^2) = \sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2$$

**Stationarity and Unconditional Variance:** If  $\alpha_1 < 1$ , the process is square integrable and stationary. Taking expectations of  $\mathbb{E}(X_t^2 \mid \mathcal{F}_{t-1}) = \alpha_0 + \alpha_1 X_{t-1}^2$  gives  $\mathbb{E}(X_t^2) = \alpha_0 + \alpha_1 \mathbb{E}(X_{t-1}^2)$ , and stationarity then yields

$$\text{Var}(X_t) = \mathbb{E}(X_t^2) = \frac{\alpha_0}{1 - \alpha_1}$$

**Key Features:**

- $(X_t)$  is a white noise:  $\text{Cov}(X_t, X_{t+h}) = \mathbb{E}\big[X_t \mathbb{E}(X_{t+h} \mid \mathcal{F}_{t+h-1})\big] = 0$  for  $h \geq 1$ , yet the  $X_t$  are *not* independent

- $X_t^2$  follows an AR(1):  $X_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2 + \nu_t$  with  $\nu_t = X_t^2 - \sigma_t^2$  a martingale difference
- For Gaussian  $\varepsilon_t$  and  $3\alpha_1^2 < 1$ , the kurtosis is  $\frac{3(1 - \alpha_1^2)}{1 - 3\alpha_1^2} > 3$ , so the tails are heavier than normal

Returns can therefore look uncorrelated while their squares are clearly autocorrelated, which is exactly what we see in real data.

### 8.3 ARCH( $p$ )

$$X_t = \sigma_t \varepsilon_t, \quad \sigma_t^2 = \alpha_0 + \sum_{i=1}^p \alpha_i X_{t-i}^2, \quad \alpha_0 > 0, \; \alpha_i \geq 0$$

The constant  $\alpha_0$  is often written  $\omega$ . The process is stationary if  $\sum_{i=1}^p \alpha_i < 1$ , in which case

$$\text{Var}(X_t) = \mathbb{E}(\sigma_t^2) = \frac{\alpha_0}{1 - \sum_{i=1}^p \alpha_i}$$

The unconditional variance is constant, while the conditional variance moves with recent squared returns. In practice many lags are needed to capture persistent volatility, which motivates GARCH.

### 8.4 GARCH(1,1)

$$X_t = \sigma_t \varepsilon_t, \quad \sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \quad \alpha_0 > 0, \; \alpha_1, \beta_1 \geq 0$$

Today’s variance is a weighted combination of a long run level, yesterday’s squared shock and yesterday’s variance.

**Stationarity:** If  $\alpha_1 + \beta_1 < 1$  then

$$\text{Var}(X_t) = \bar{\sigma}^2 = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}$$

The sum  $\alpha_1 + \beta_1$  is the **persistence**; estimates on daily equity returns are typically close to one.

**ARMA Representation of Squares:** With  $\nu_t = X_t^2 - \sigma_t^2$ ,

$$X_t^2 = \alpha_0 + (\alpha_1 + \beta_1) X_{t-1}^2 + \nu_t - \beta_1 \nu_{t-1}$$

so  $X_t^2$  behaves like an ARMA(1,1). Substituting recursively shows that GARCH(1,1) is an ARCH( $\infty$ ) with geometrically decaying weights  $\alpha_1 \beta_1^{k-1}$ , which explains its parsimony.

**Variance Forecasts:** For  $h \geq 1$ ,

$$\mathbb{E}(\sigma_{t+h}^2 \mid \mathcal{F}_t) = \bar{\sigma}^2 + (\alpha_1 + \beta_1)^{h-1} (\sigma_{t+1}^2 - \bar{\sigma}^2)$$

Forecasts revert geometrically to the long run variance at rate  $\alpha_1 + \beta_1$ .

### 8.5 GARCH( $p, q$ )

$$X_t = \sigma_t \varepsilon_t, \quad \sigma_t^2 = \alpha_0 + \sum_{i=1}^p \alpha_i X_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2$$

with  $\alpha_0 > 0$  and all other coefficients nonnegative. It is (weakly) stationary if  $\sum_i \alpha_i + \sum_j \beta_j < 1$ , and then

$$\text{Var}(X_t) = \frac{\alpha_0}{1 - \sum_{i=1}^p \alpha_i - \sum_{j=1}^q \beta_j}$$

Conventions for which index belongs to which sum differ between texts and software, so it is worth checking before reading off estimates.

**In Practice:** A typical workflow fits a mean model (often a constant or low order ARMA) to returns, checks the squared residuals for remaining autocorrelation with a Ljung Box test, fits a GARCH to the residuals by (quasi) maximum likelihood, and validates by testing the standardised residuals  $\hat{\varepsilon}_t = X_t / \hat{\sigma}_t$  and their squares for white noise. Extensions such as EGARCH, GJR GARCH and GARCH in mean allow for asymmetric responses to good and bad news and for a risk premium in the mean.